KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2016
Department of Computer Science and Engineering
CSE 4105
Computer Networks

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

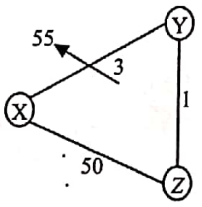
(Answer **ANY THREE** questions from this section in Script A)

1. a) What is layering in computer networks? Write down the major functions of network layer. (07)
   b) What are the advantages of packet switching over circuit switching? (06)
   c) Suppose a TCP message that contains 2048 bytes of data and 20 bytes of TCP header is passed to IP for delivery across two networks of the Internet (i.e., from the source host to a router to the destination host). The first network uses 14-bytes headers and has an MTU of 1024 bytes; the second user 8-byte header with an MTU of 512 bytes. Give the sizes and offsets of the sequence of fragments delivered to the network layer at the destination host. Assume all IP header are 20 bytes. (12)
   d) Differentiate between IPv4 and IPv6 header format. Explain the technologies that are used for transition from IPv4 to IPv6. (10)

2. a) Write down the pros and cons of static and dynamic routing. Explain link-state routing algorithm with a proper example. (13)
   b) What is DHCP? How does DHCP work? Write down the advantages and disadvantages of DHCP. (12)
   c) What does ICMP stand for? What does it do? Describe the error reporting messages of ICMP. (10)

3. a) "A transport-layer protocol provides logical communication between processes running on different hosts, a network-layer protocol provides logical communication between hosts."- Justify the statement. (08)
   b) What are the essential tools of RDT protocol? Describe the working mechanism of RDT 2.0. How can the performance problem of RDT be solved? (13)
   c) Describe the TCP timers in Brief. (08)
   d) Since UDP is unreliable, why is it used at transport layer? (06)

4. a) Consider the following graph. Assume that link (x,y) cost has been changed from 3 to 55. Show that the poissoned reverse technique can resolve the count to infinity problem in this case. (11)



   b) In theory TCP is fair, but in face it is not. Give an example. (07)
   c) What is the NAT traversal problem for P2P applications, say Skype? (05)
   d) Suppose the values of p and q are 7 and 13 respectively. Find suitable public and private keys for RSA algorithm. Verify the keys for the message "CSE". (12)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What are the techniques to resolve IPv4 address space problem? Explain in Brief. (10)
   b) What is digital signature? How authentication is provided using digital signature. (10)
   c) What is Diffie Hellman key exchange algorithm? Explain its working principle with an example. (10)
   d) What do firewalls do? Explain shortly. (05)

6. a) "Sockets are the software interface"-Justify the statement. (05)
   b) What are the transport services does an application need? Briefly describe them. (10)
   c) What is the benefit of stateless property of HTTP server? How HTTP servers identify users with stateless property? Explain briefly. (10)
   d) Differentiate GET and Conditional GET commands of HTTP. How does conditional GET command improve performance of web proxy and local cash mechanism? (10)

7. a) Why is it said that FTP sends control information "out of band"? (05)
   b) Suppose Alice, with a web-based email account (such as gmail) sends a message to Bob, who accesses his mail from his mail server using POP3. Discuss how the message gets from Alice's host to Bob's host. Be sure to list the series of application layer protocols that are used to move the message between the two hosts. (10)
   c) Write filtering/forwarding algorithm of a switch. List the actions with corresponding condition(s) that a switch may perform when it receives a frame. (08)
   d) What is service provided by DNS system? Explain load distribution briefly. (06)
   e) If all the links in the Internet were to provide reliable delivery service, would the TCP reliable delivery service be redundant? Why or why not? (06)

8. a) Consider an e-commerce site that wants to keep a purchase record for each of its customer. Describe how this can be done with cookies? (08)
   b) Why are content distribution networks (CDNs) important in internet system? Describe the role of DNS in CDN. (07)
   c) Write down the advantages and disadvantages of polling and Token passing MAC protocol. (09)
   d) Discuss Byte stuffing with an appropriate example. (06)
   e) Why a network interface card is masked as semi-autonomous device? (05)